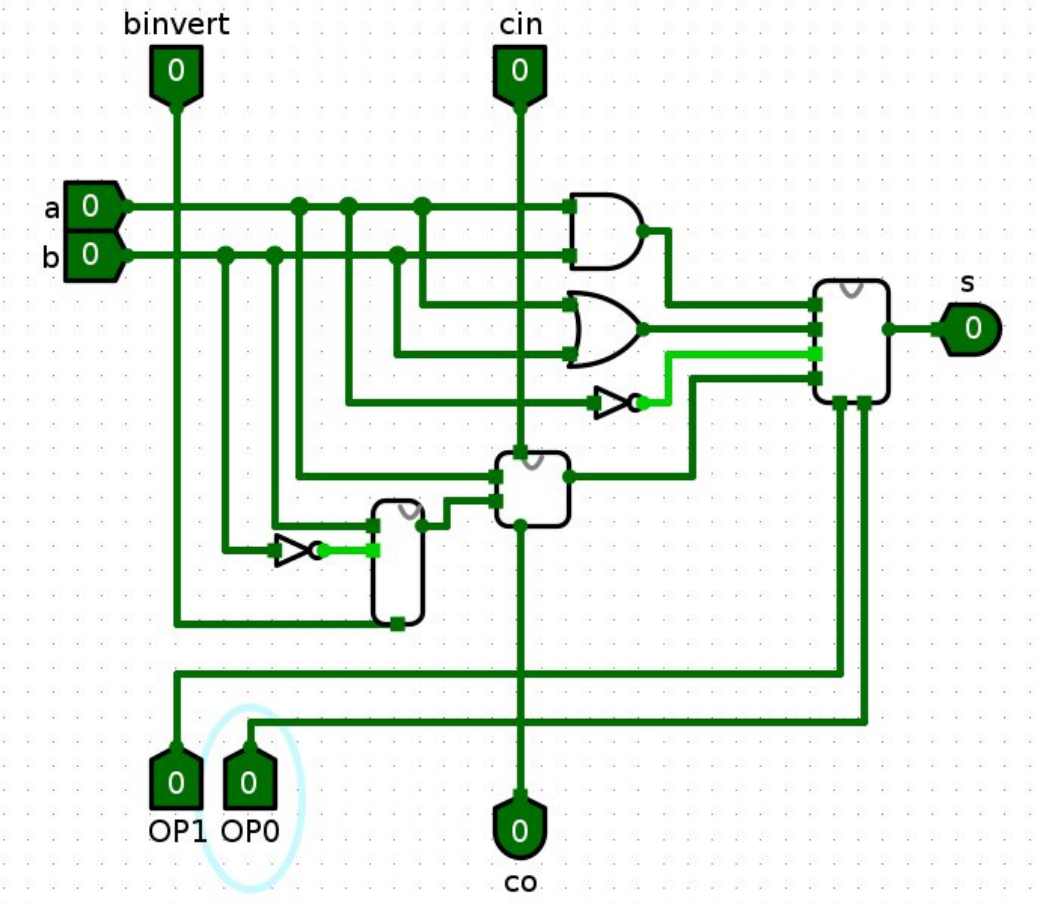
Arquitetura de Computadores 2 – Exercício Prático 02

Samuel Fragoso - 902665

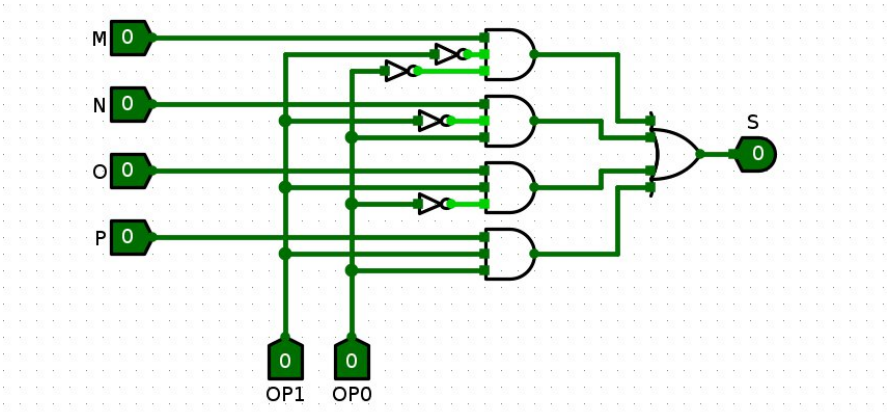
PARTE 1

INSTRUÇÃO REALIZADA	BINÁRIO (A,B,Op.code)	Valor em Hexa (0x....)	Resultado em binário
AND(A,B)	0010 0001 00	(0000 1000 0100) = 0x084	0000
OR(A,B)	0010 0011 01	0x08D	0011
SOMA(A,B)	0010 0011 11	0x08F	0101
NOT(A)	1100 0011 10	0x30E	0011
AND(B,A)	1100 1101 00	0x334	1100

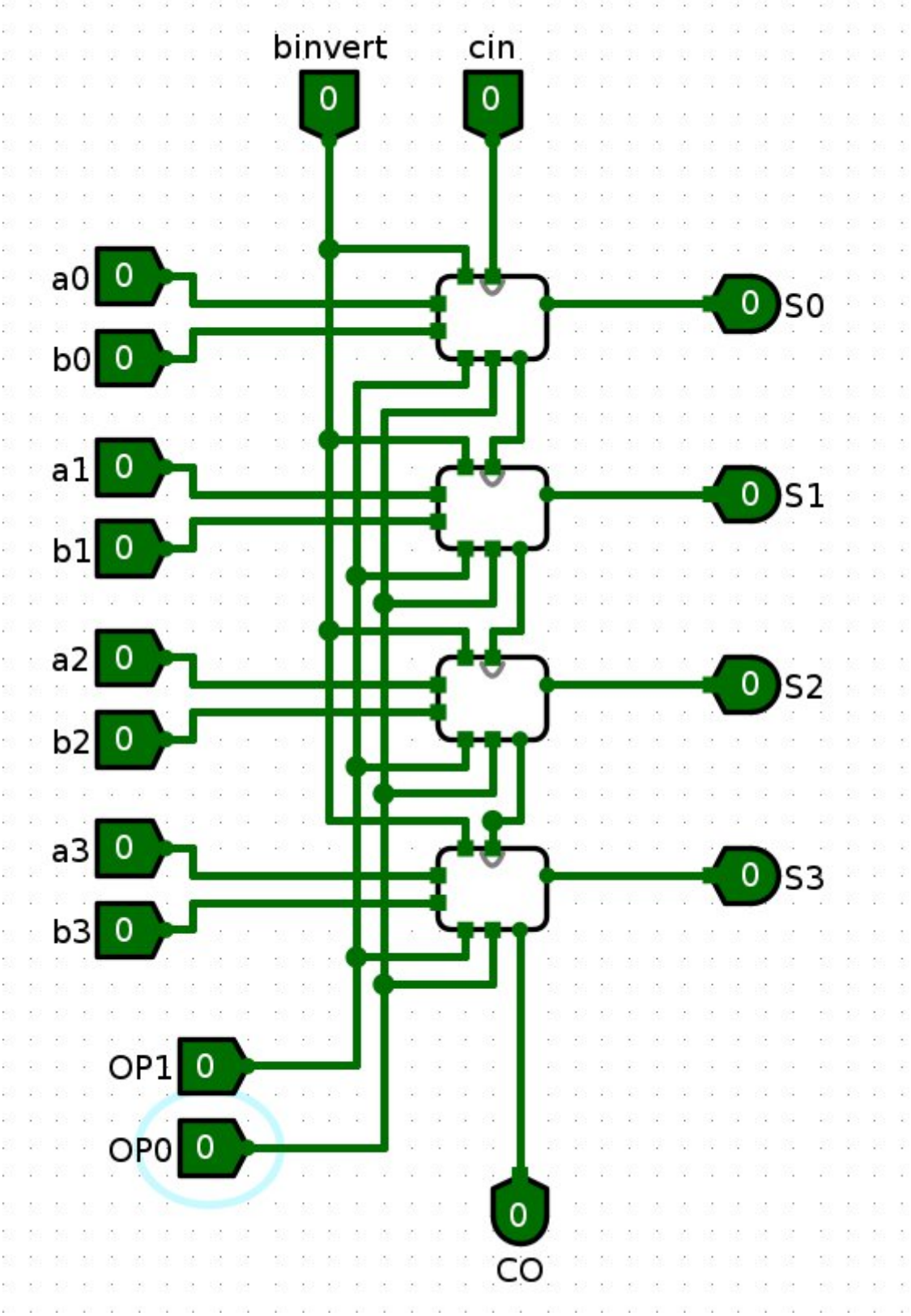
ULA 1 bit



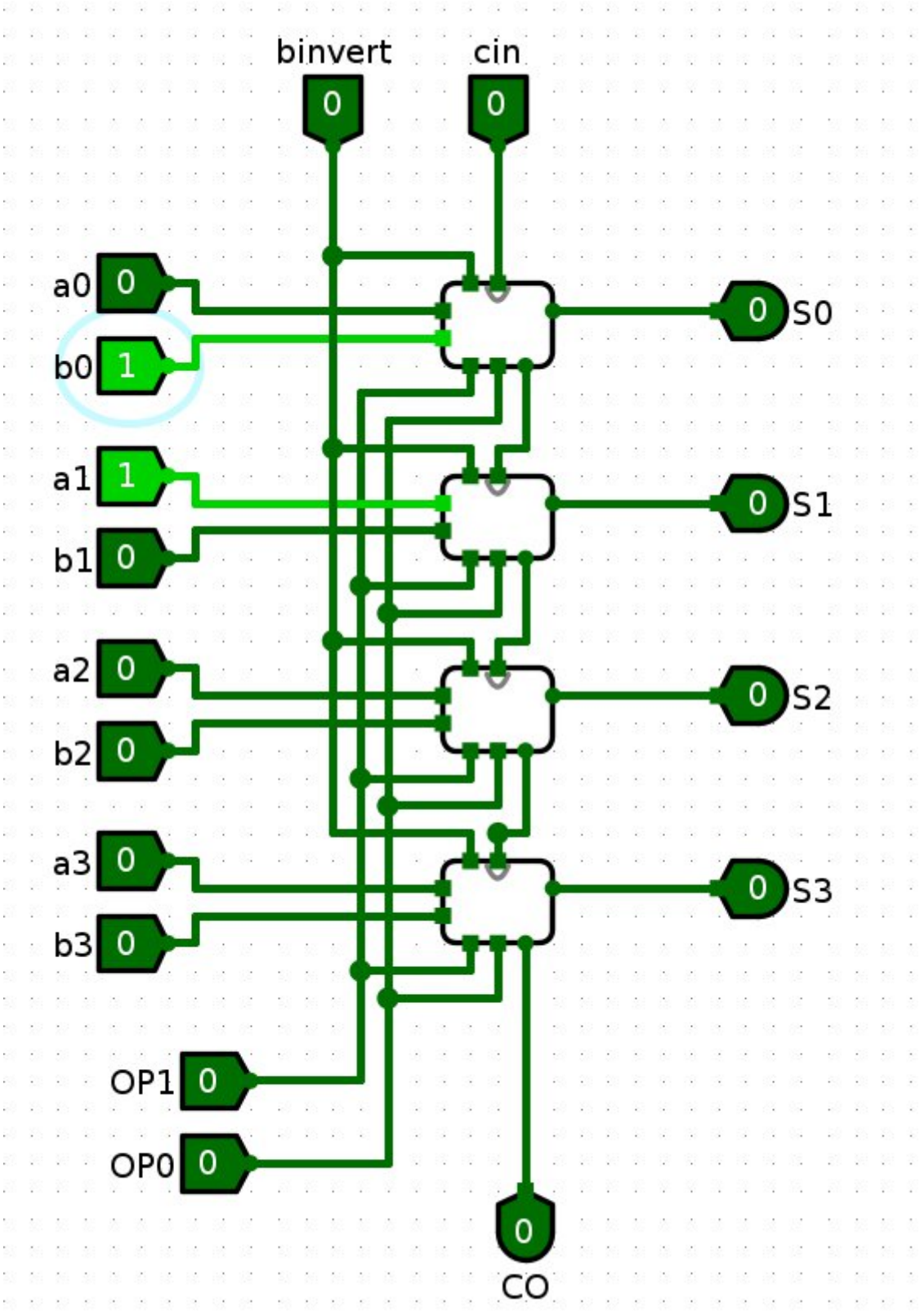
MUX



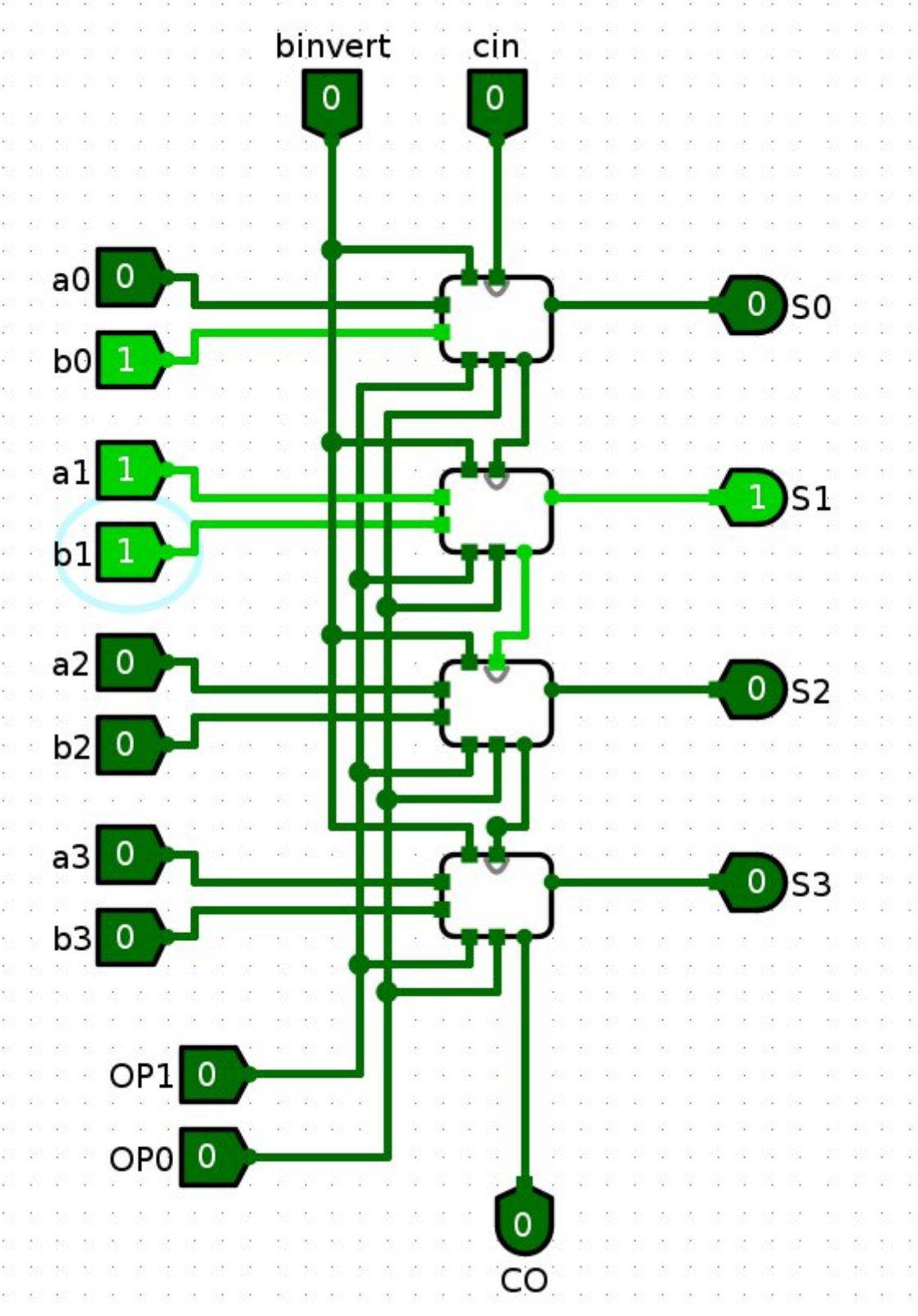
ULA 4 bits


binvert
cin
0
0
a0
0
b0
0
a1
0
b1
0
a2
0
b2
0
a3
0
b3
0
OP1
0
OP0
0
0
S0
0
S1
0
S2
0
S3
0
CO

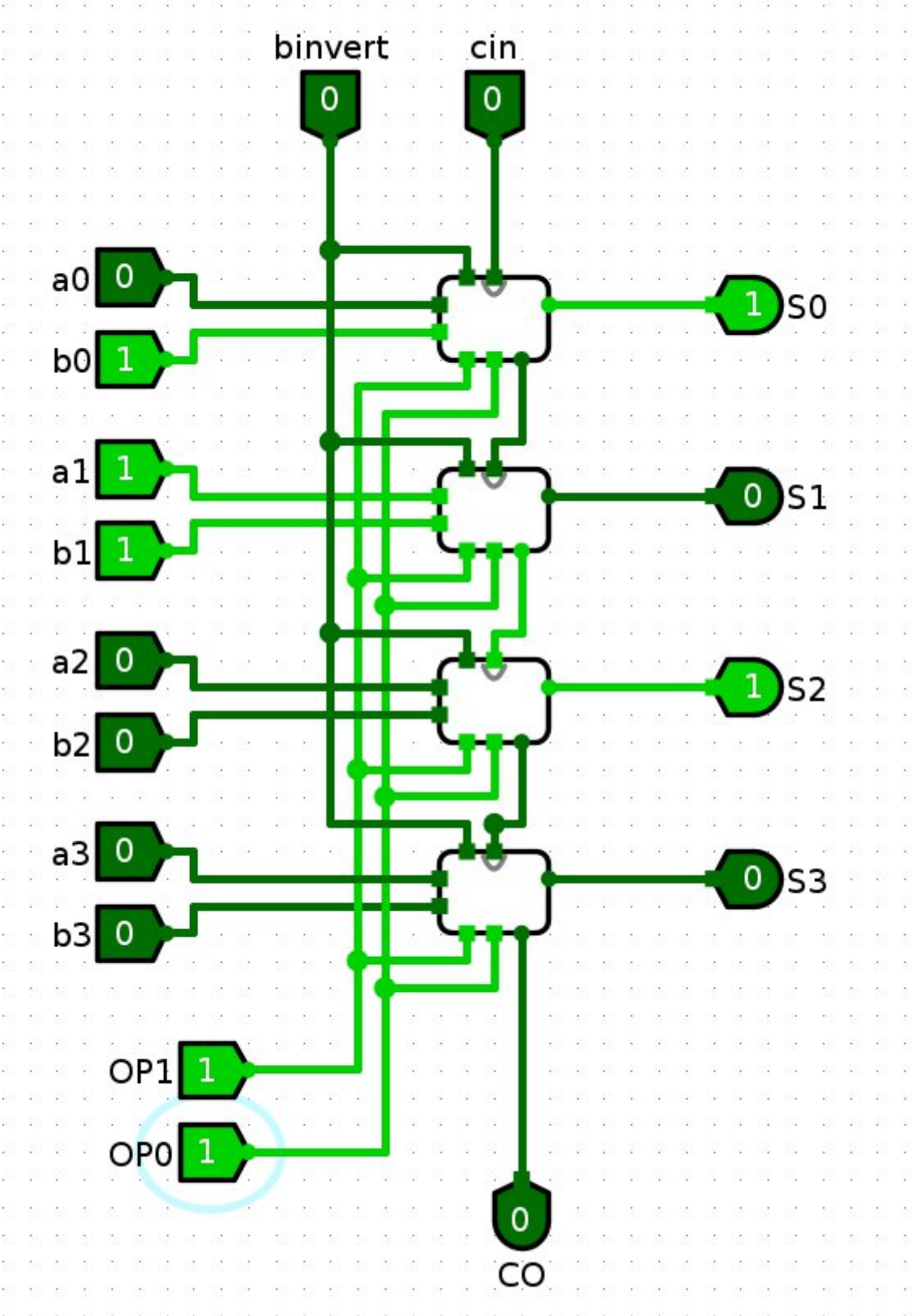
00(AND)


binvert
cin
0
0
a0
0
b0
1
a1
1
b1
0
a2
0
b2
0
a3
0
b3
0
OP1
0
OP0
0
0
S0
0
S1
0
S2
0
S3
0
CO

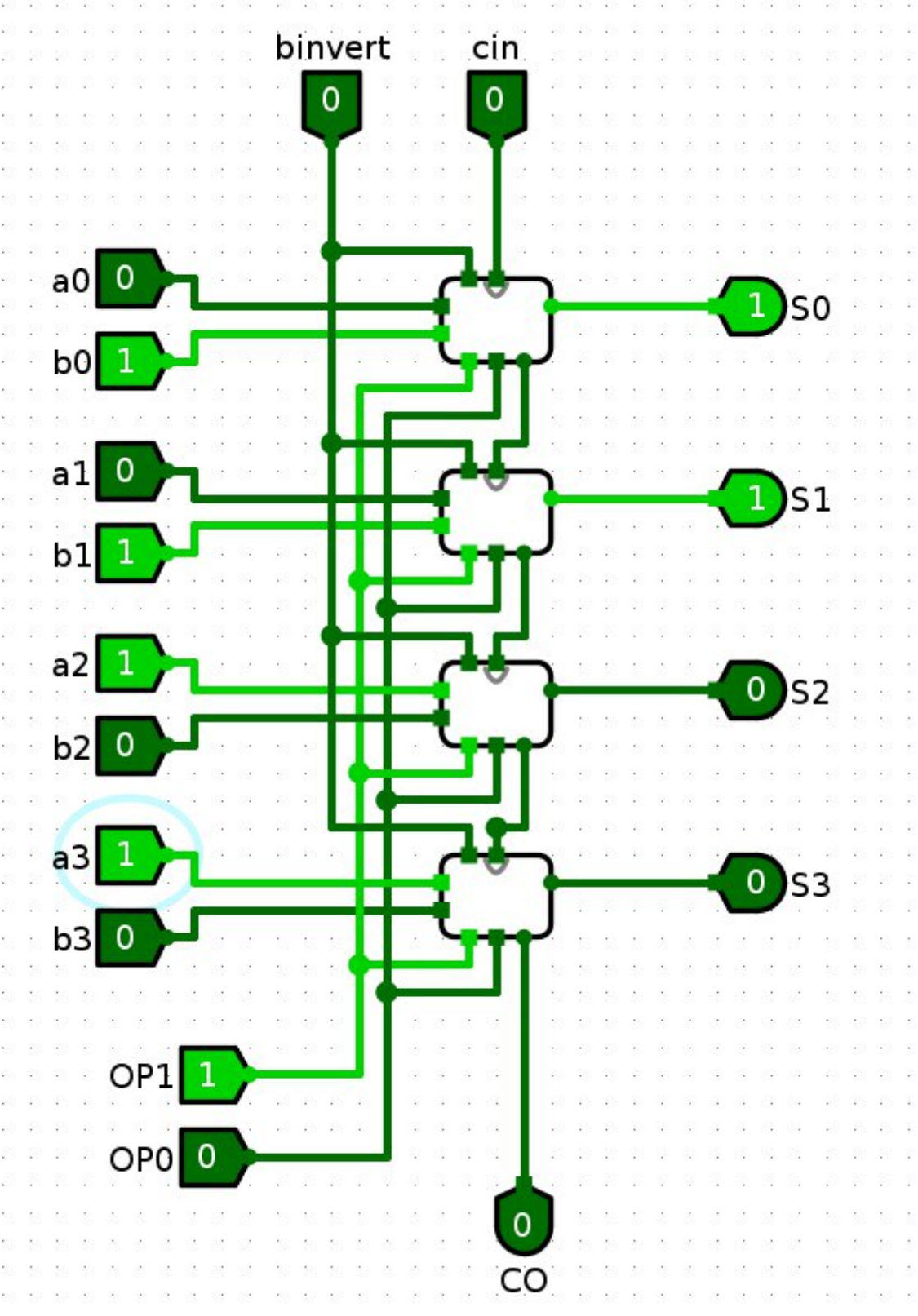
01(OR)


binvert
cin
0
0
a0
0
b0
1
a1
1
b1
1
a2
0
b2
0
a3
0
b3
0
OP1
0
OP0
0
0
S0
1
S1
0
S2
0
S3
0
CO

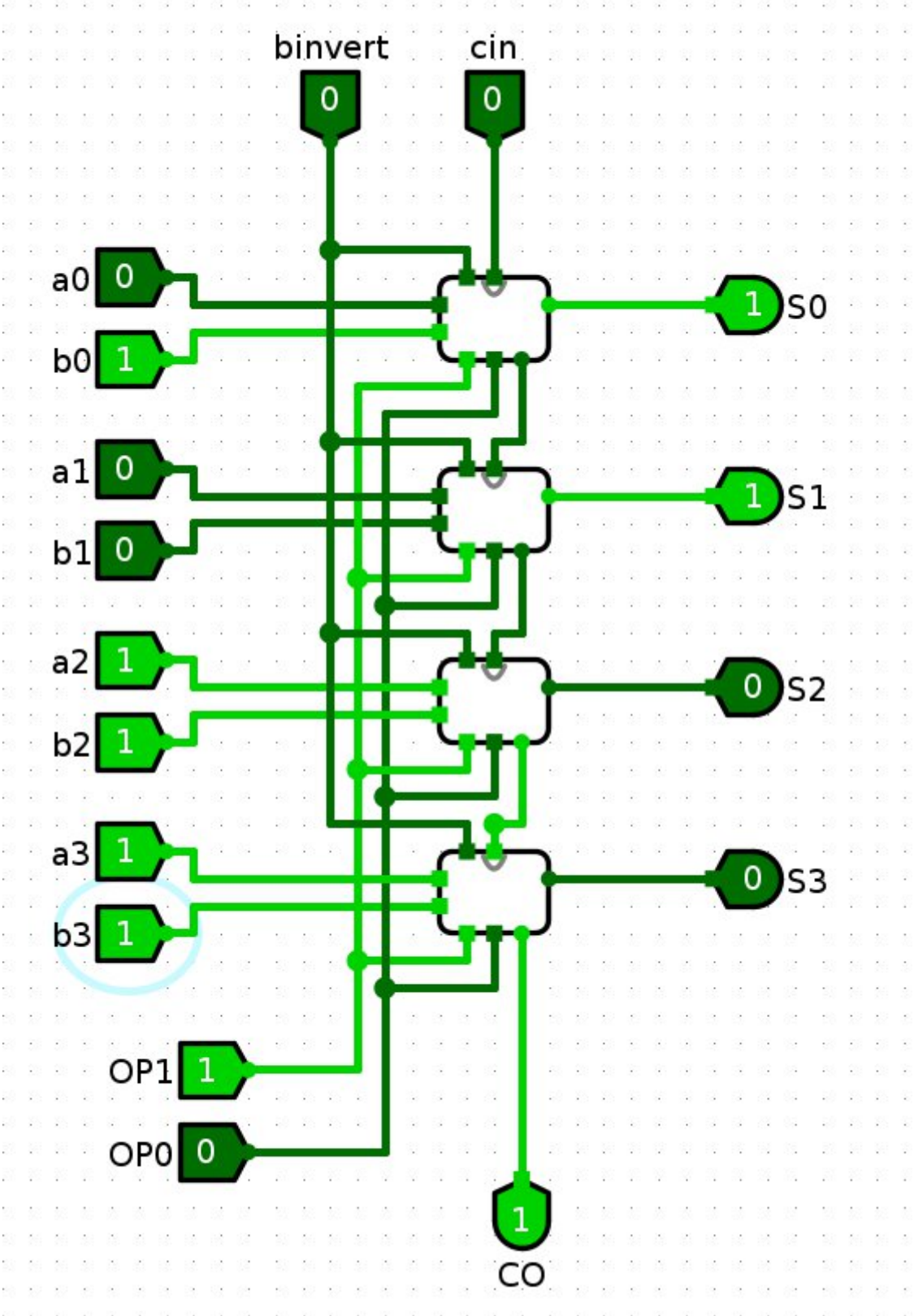
11(SOMA)


binvert
cin
0
0
a0
0
b0
1
a1
1
b1
1
a2
0
b2
0
a3
0
b3
0
OP1
1
OP0
1
1
S0
0
S1
1
S2
0
S3
0
CO

10(NOT)

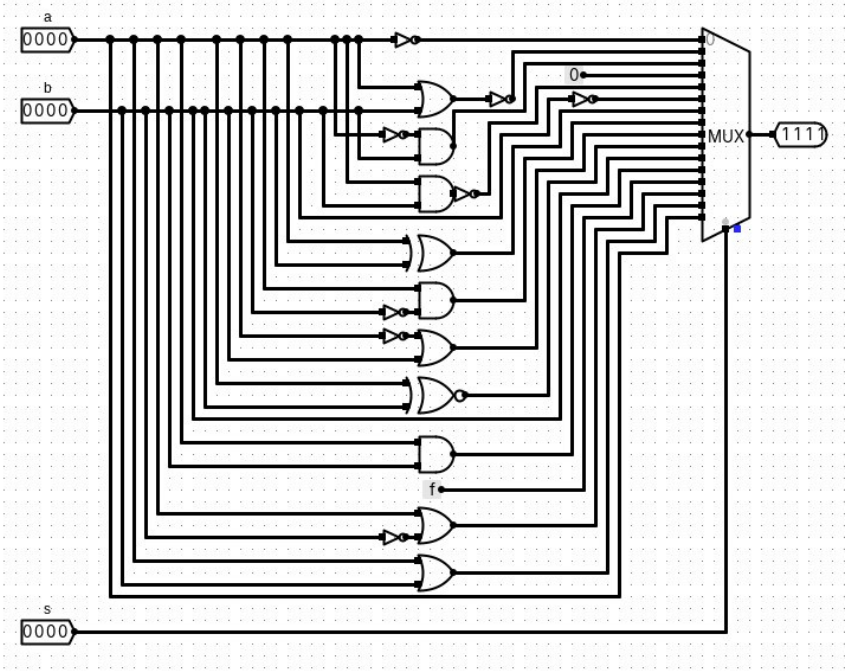

binvert
cin
0
0
a0
0
b0
1
a1
0
b1
1
a2
1
b2
0
a3
1
b3
0
OP1
1
OP0
0
1
S0
1
S1
0
S2
0
S3
0
CO

00 (AND, calcula B,A)



PARTE 2

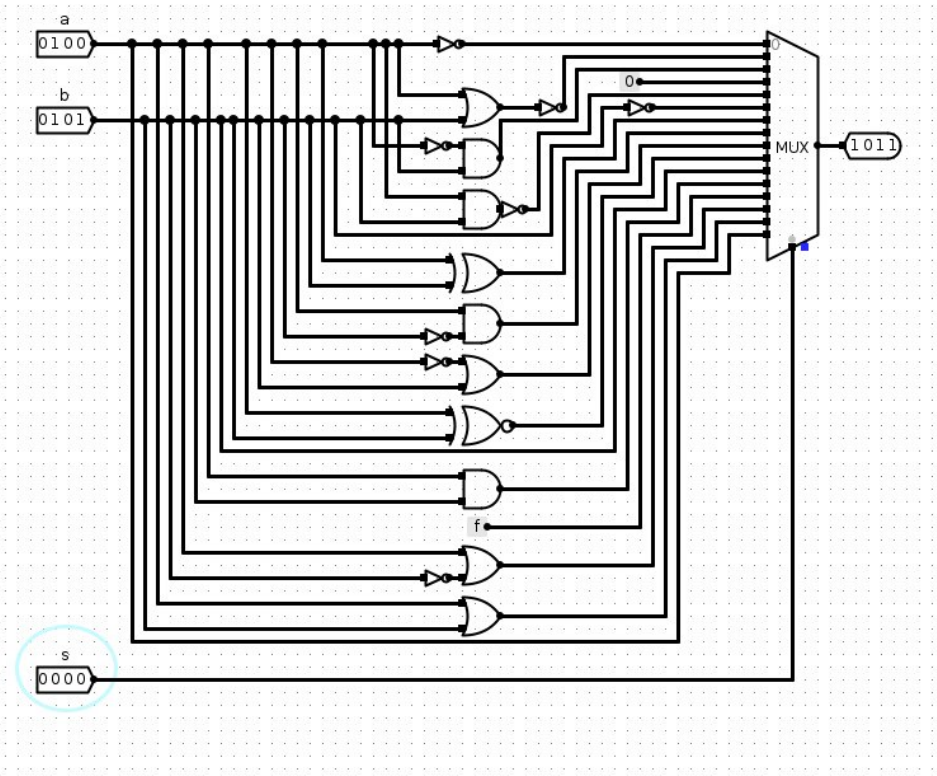
Instruções	Binário	Resultado da operação
450	0100 0101 0000	B
CB1	1100 1011 0001	7
A32	1010 0011 0010	1
C43	1100 0100 0011	0
124	0001 0010 0100	F
785	0111 1000 0101	7
9B6	1001 1011 0110	2
CD7	1100 1101 0111	0
FE8	1111 1110 1000	E
649	0110 0100 1001	D
D9A	1101 1001 1010	9
FCB	1111 1100 1011	C
63C	0110 0011 1100	F
98D	1001 1000 1101	F
76E	0111 0110 1110	7
23F	0010 0011 1111	2



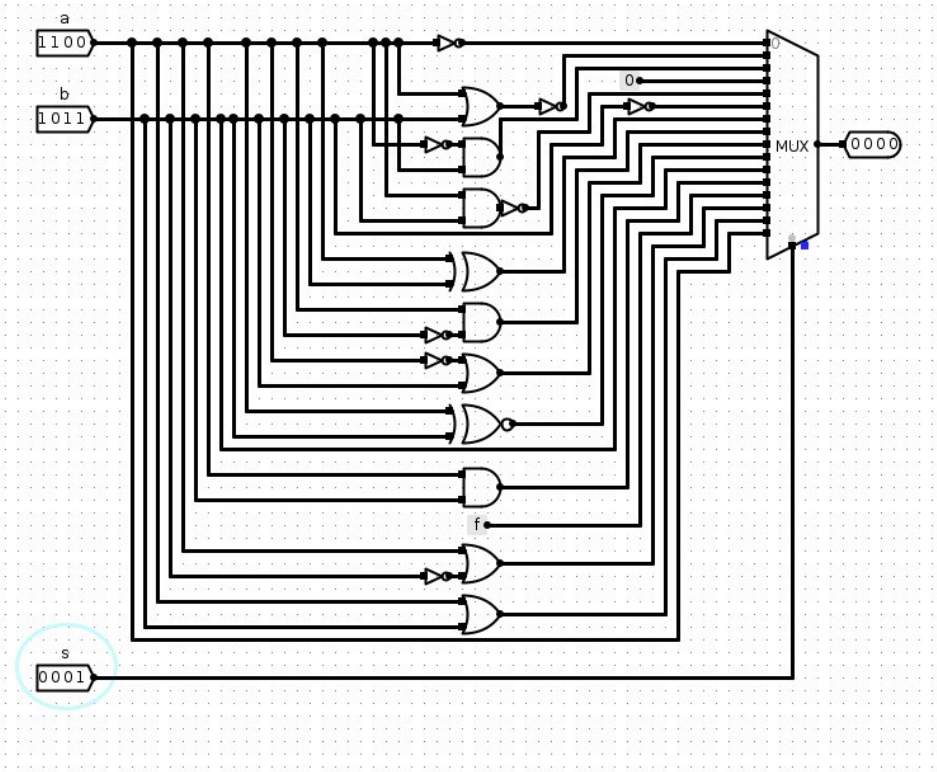
Se o objetivo fosse realmente testar esta ULA, quantas linhas a nossa tabela verdade deveria ter ou seja, a tabela que você preencheu deveria ter quantas linhas?

R: A tabela verdade completa deveria possuir 4096 linhas, pois existem 12 entradas binárias (4 bits para A, 4 bits para B e 4 bits para seleção), resultando em $2^{12} = 4096$ combinações possíveis.

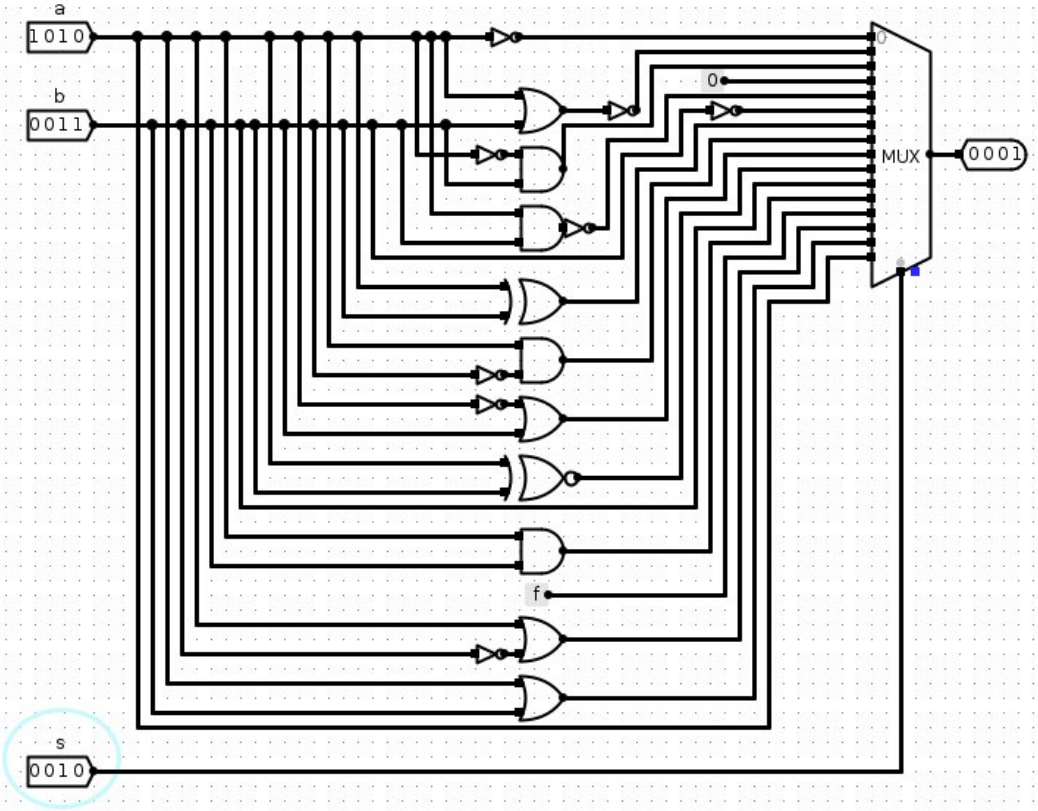
INSTRUÇÃO 450



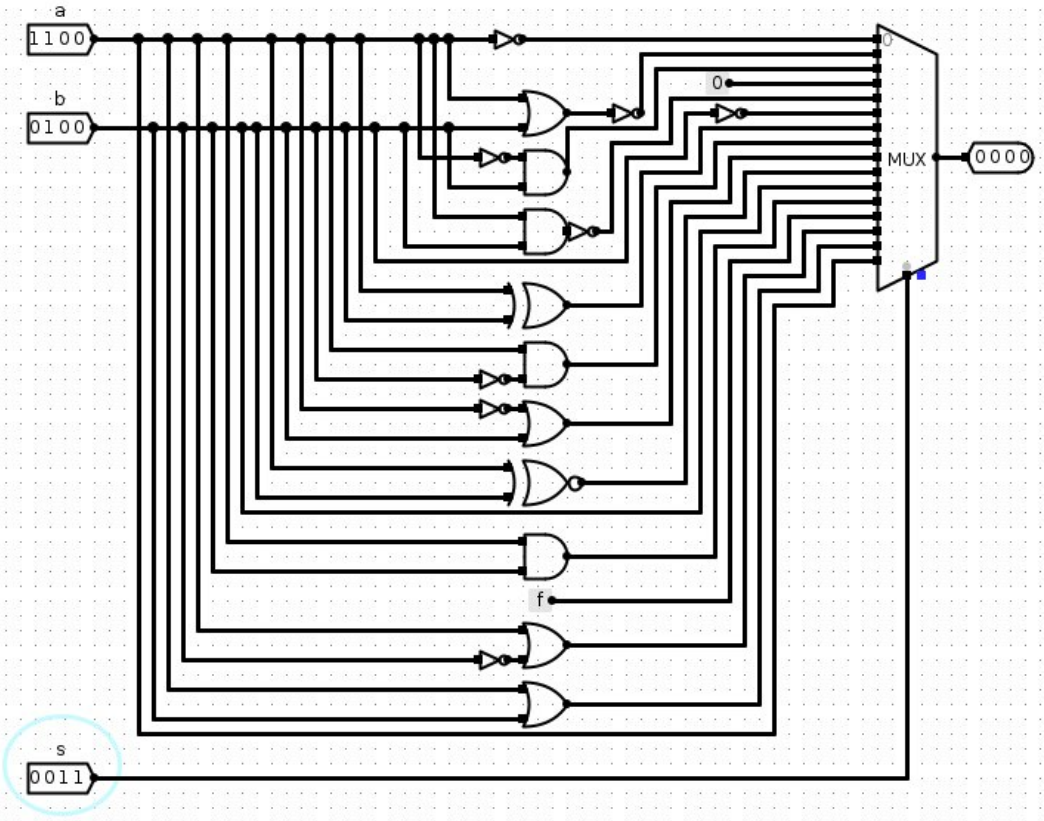
INSTRUÇÃO CB1



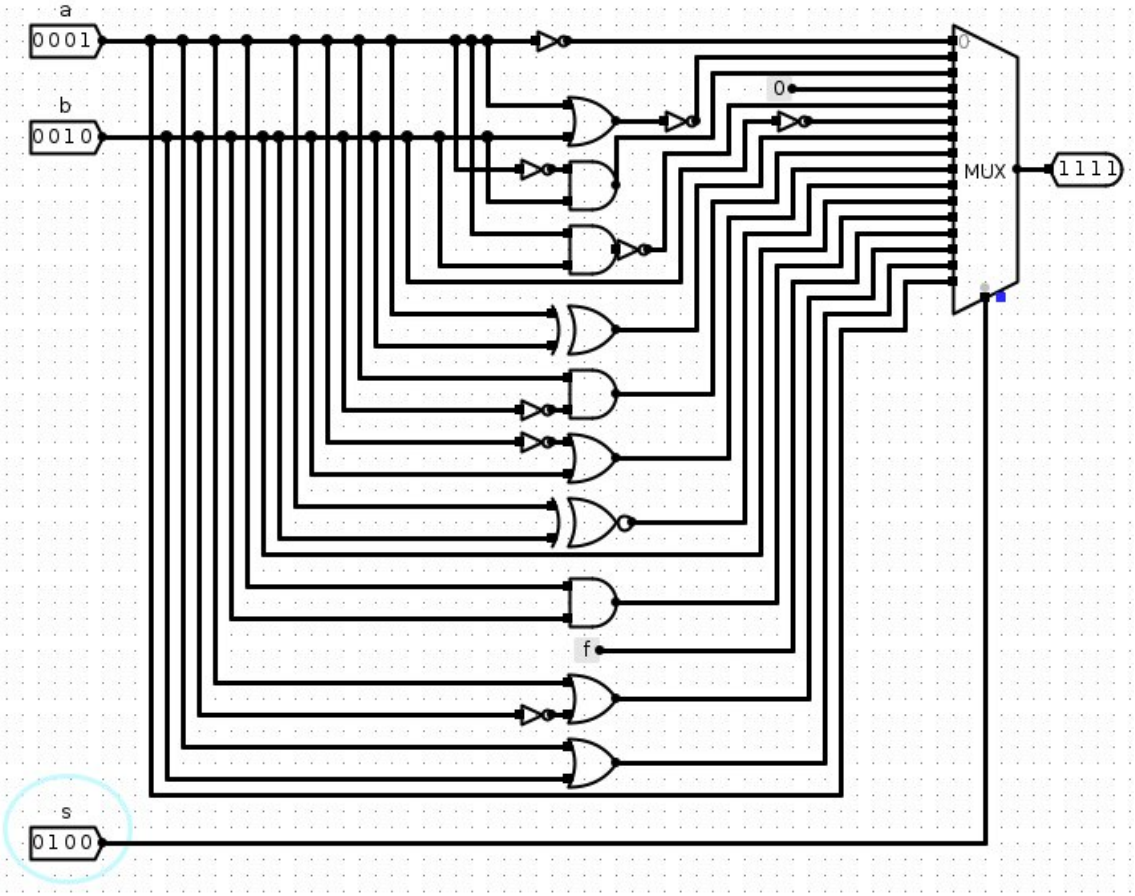
INSTRUÇÃO A32



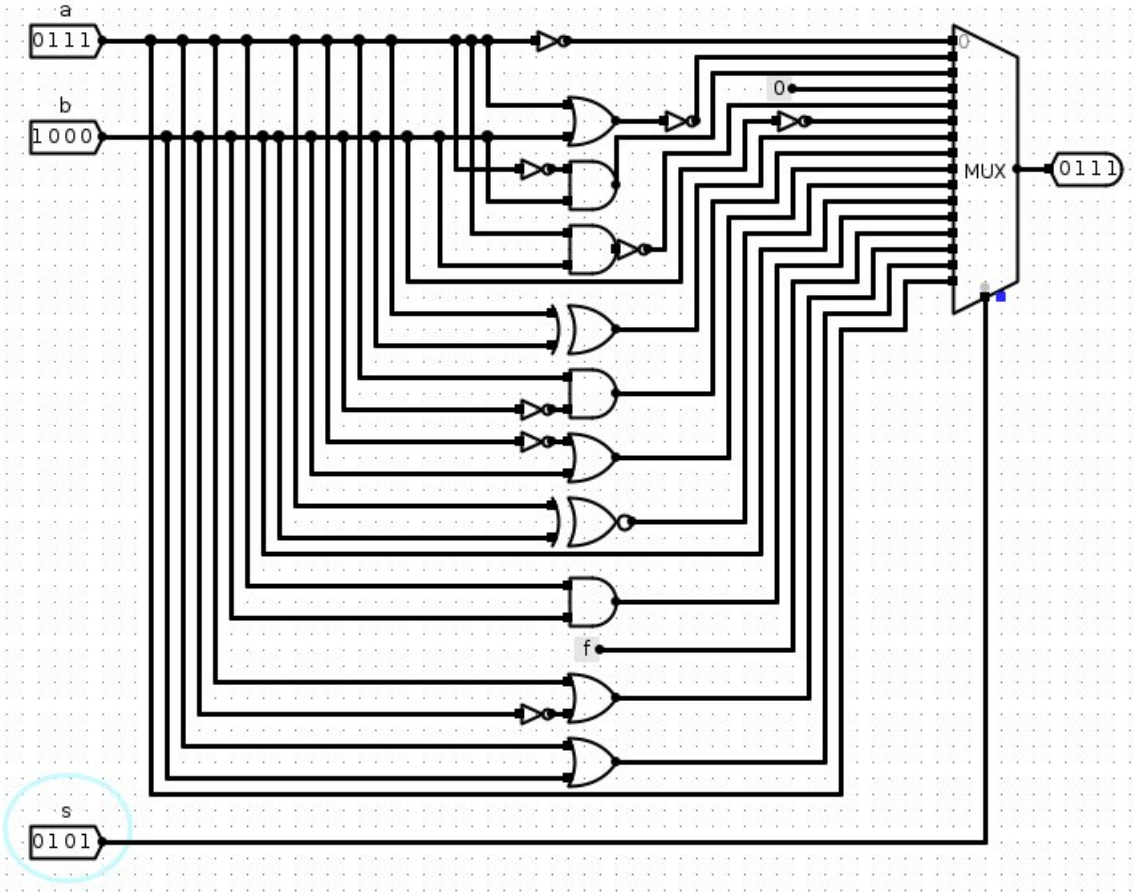
INSTRUÇÃO C43



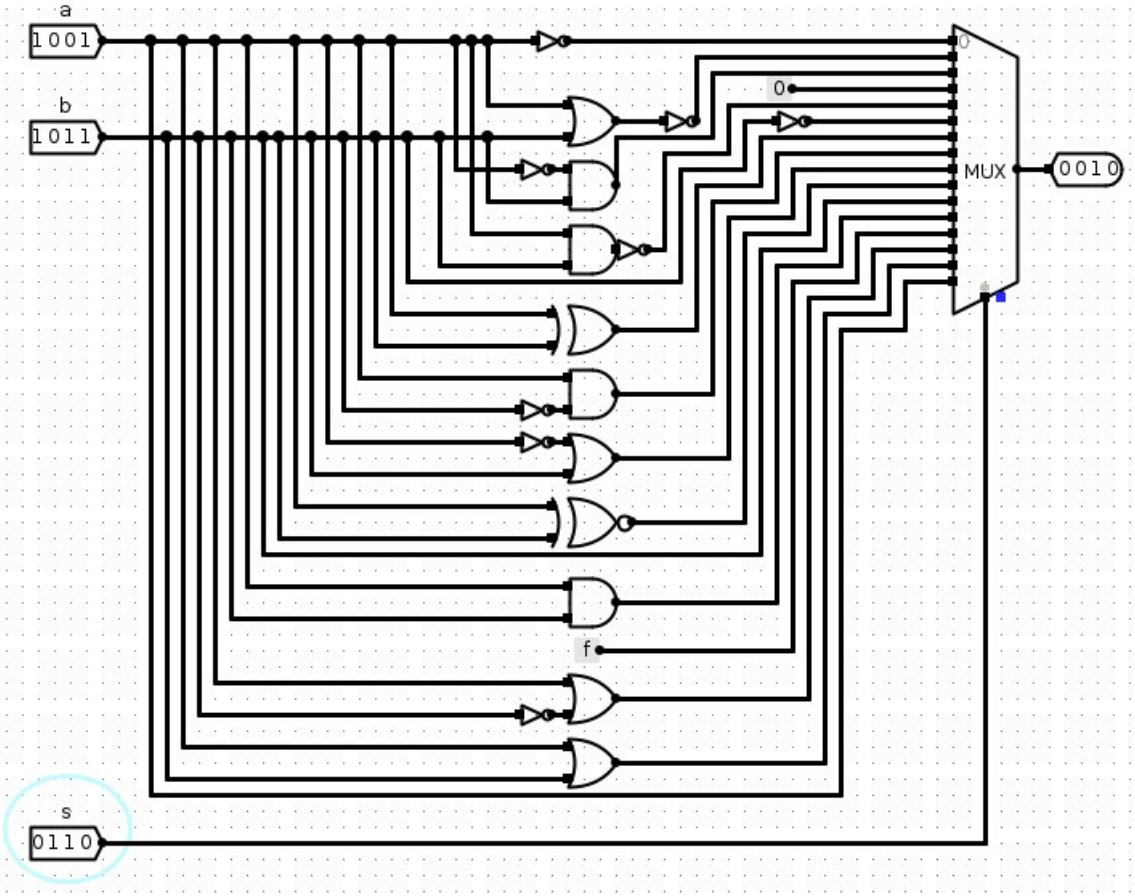
INSTRUÇÃO 124



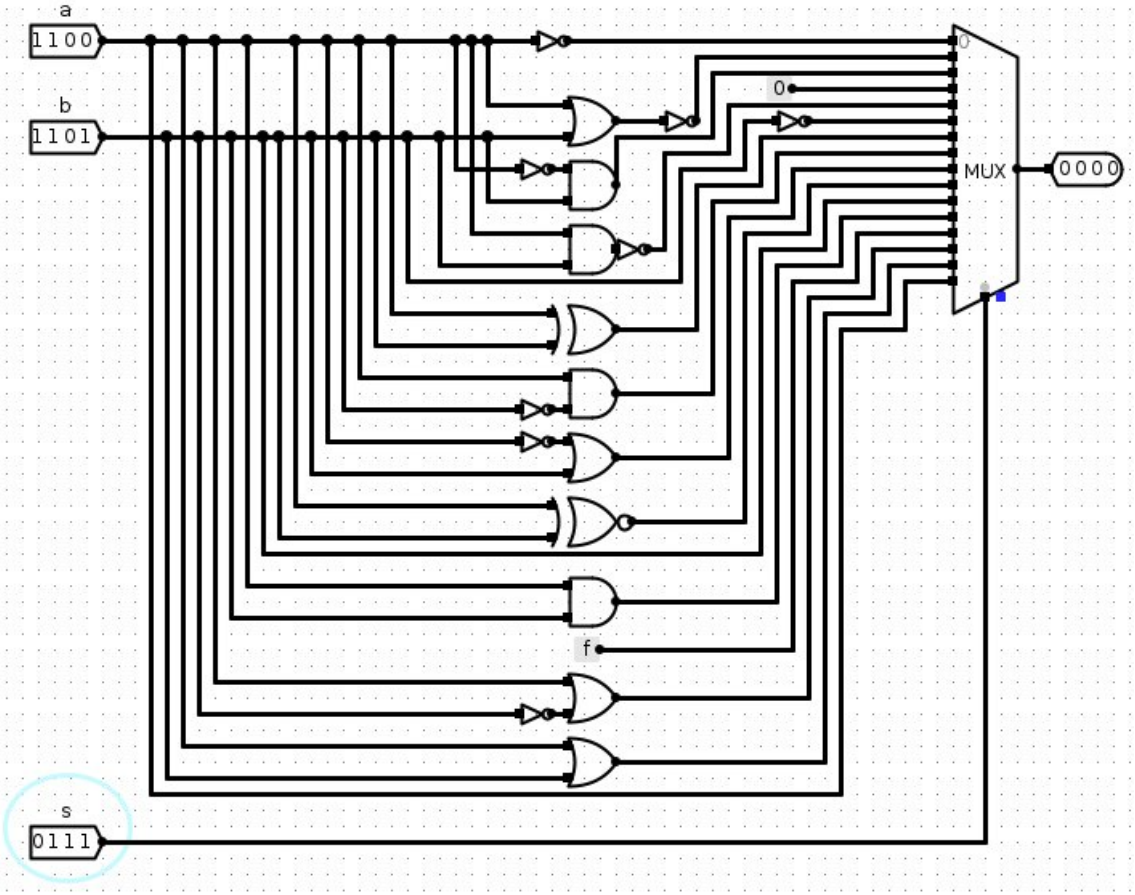
INSTRUÇÃO 785



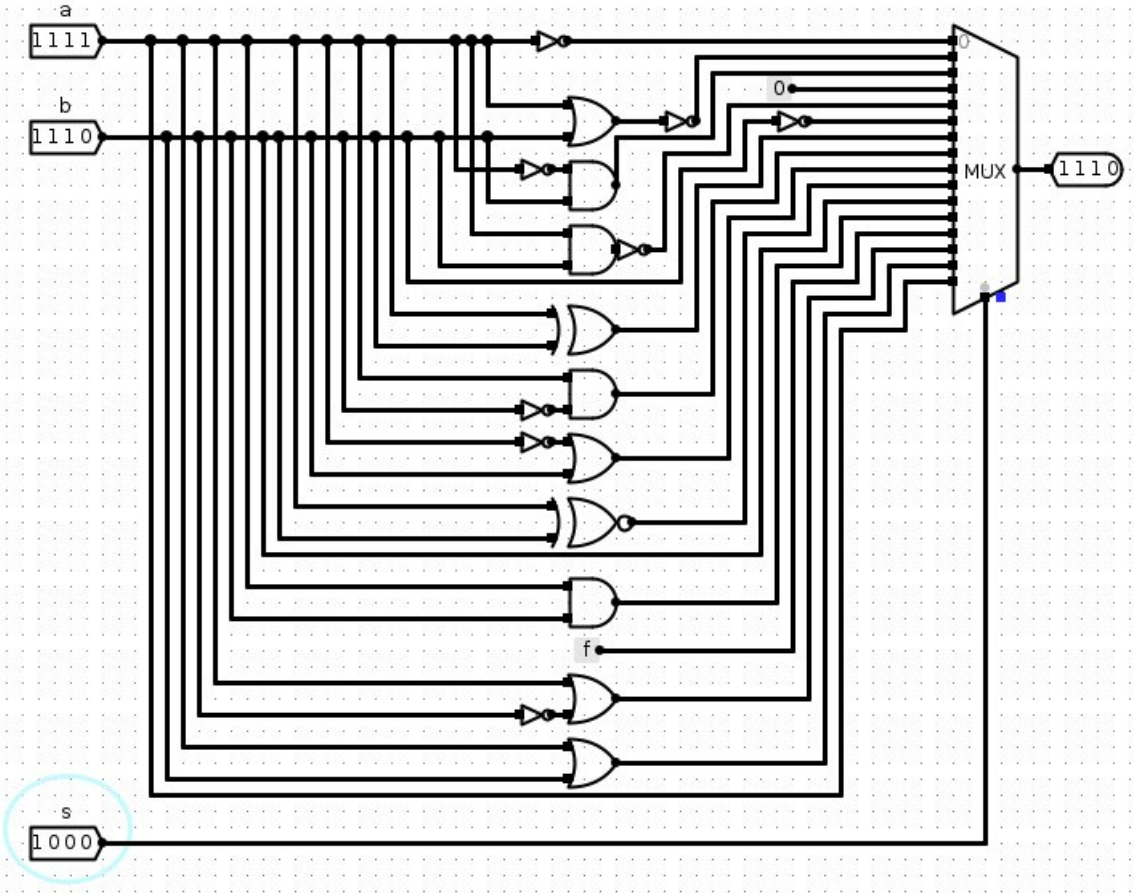
INSTRUÇÃO 9B6



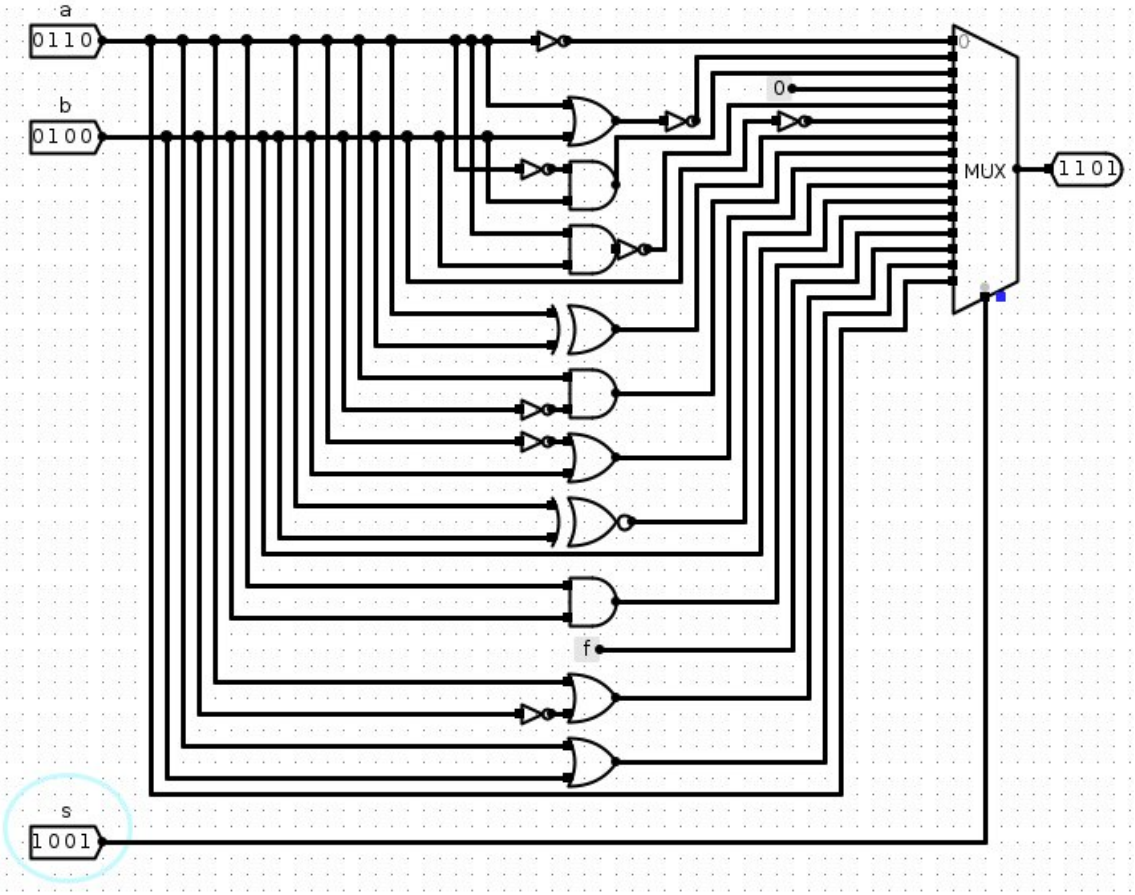
INSTRUÇÃO CD7



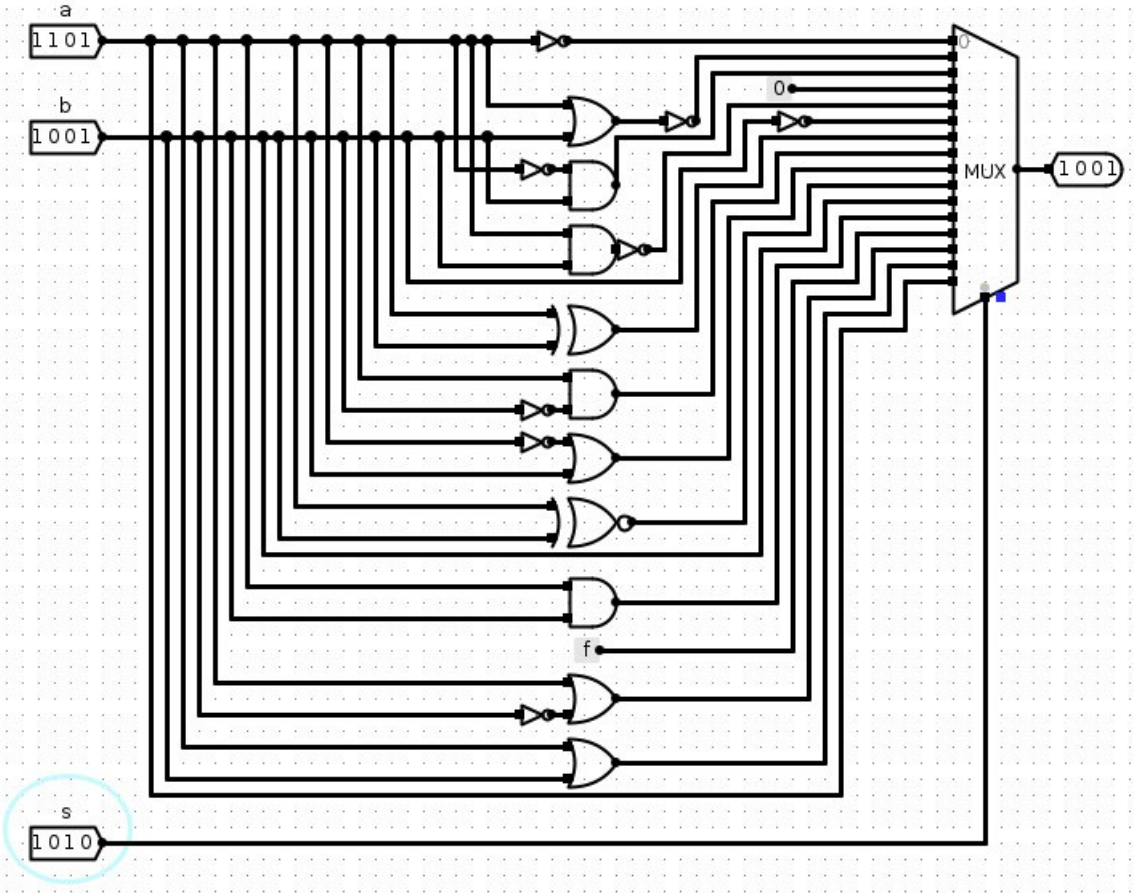
INSTRUÇÃO FE8



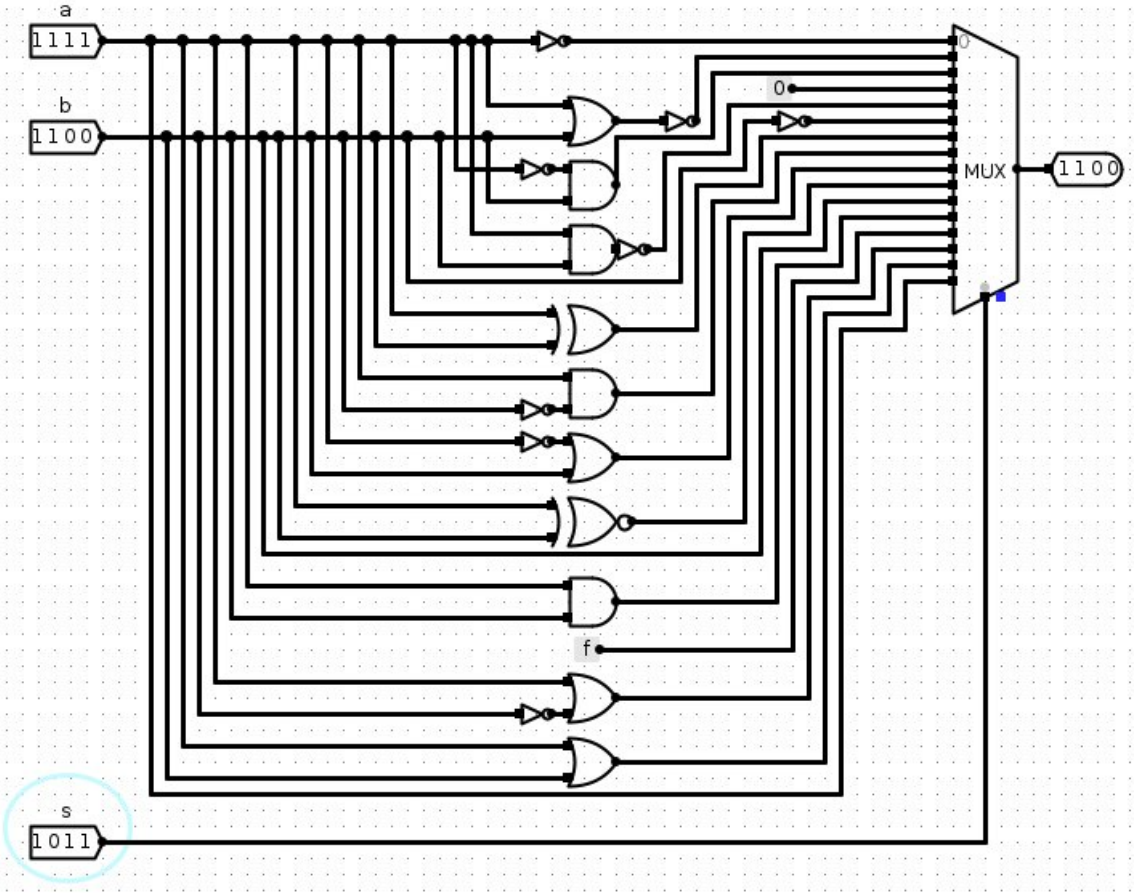
INSTRUÇÃO 649



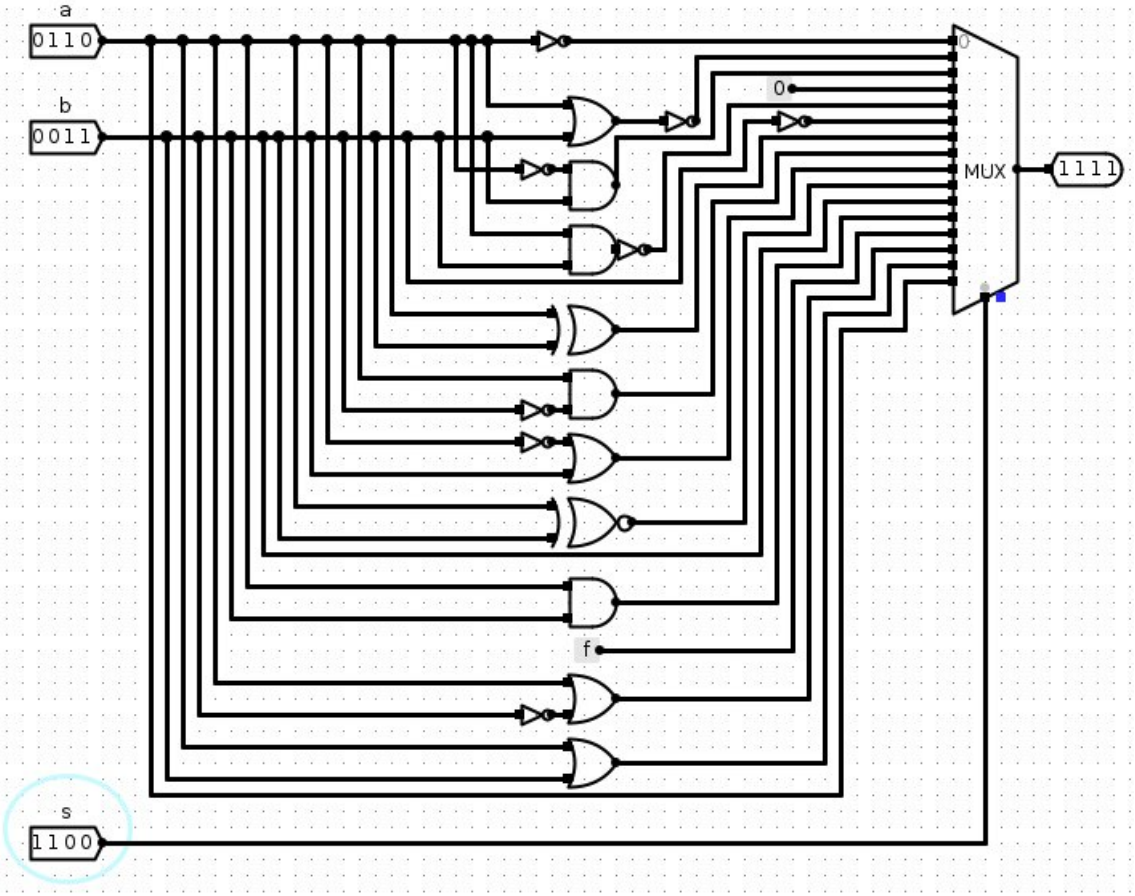
INSTRUÇÃO D9A



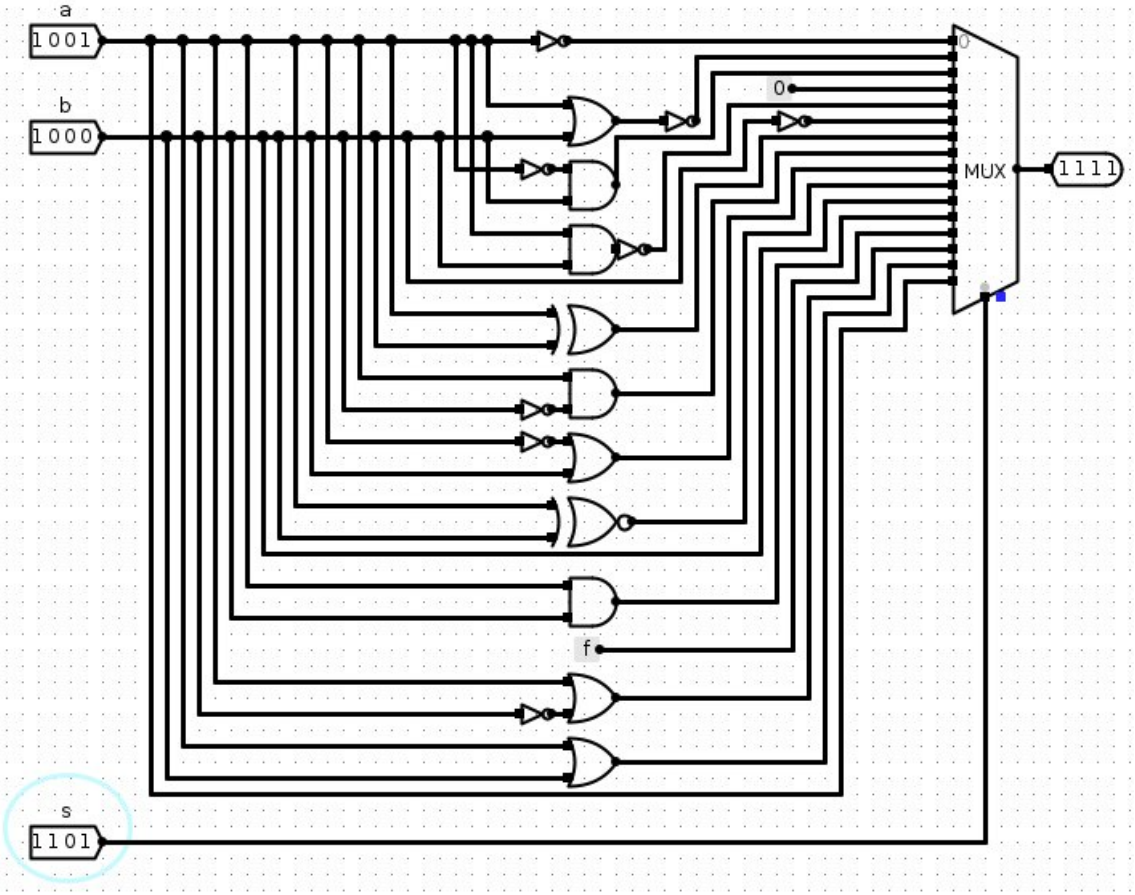
INSTRUÇÃO FCB



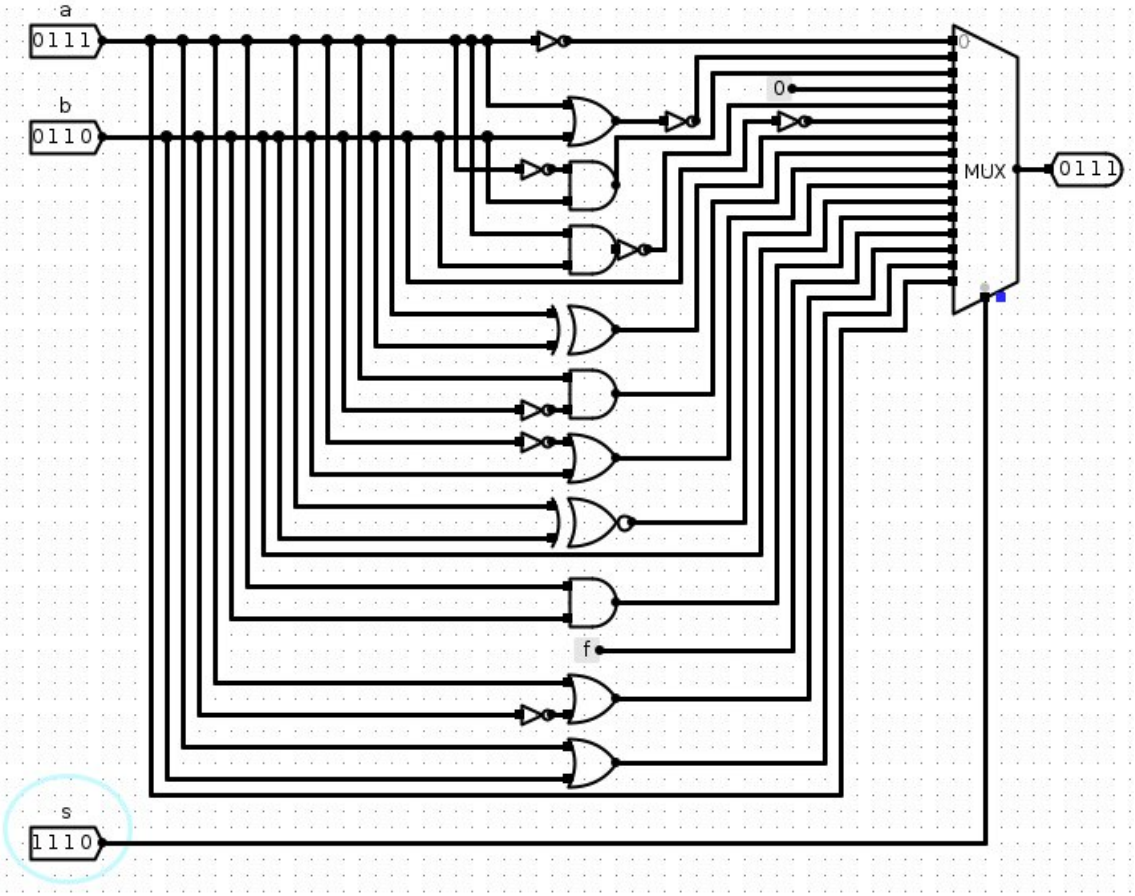
INSTRUÇÃO 63C



INSTRUÇÃO 98D



INSTRUÇÃO 76E



INSTRUÇÃO 23F

